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Antibiotic resistance pattern and molecular detection of ESBL-associated genes in *E. coli* from surface and wastewater of Islamabad capital territory, Pakistan

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ABSTRACT

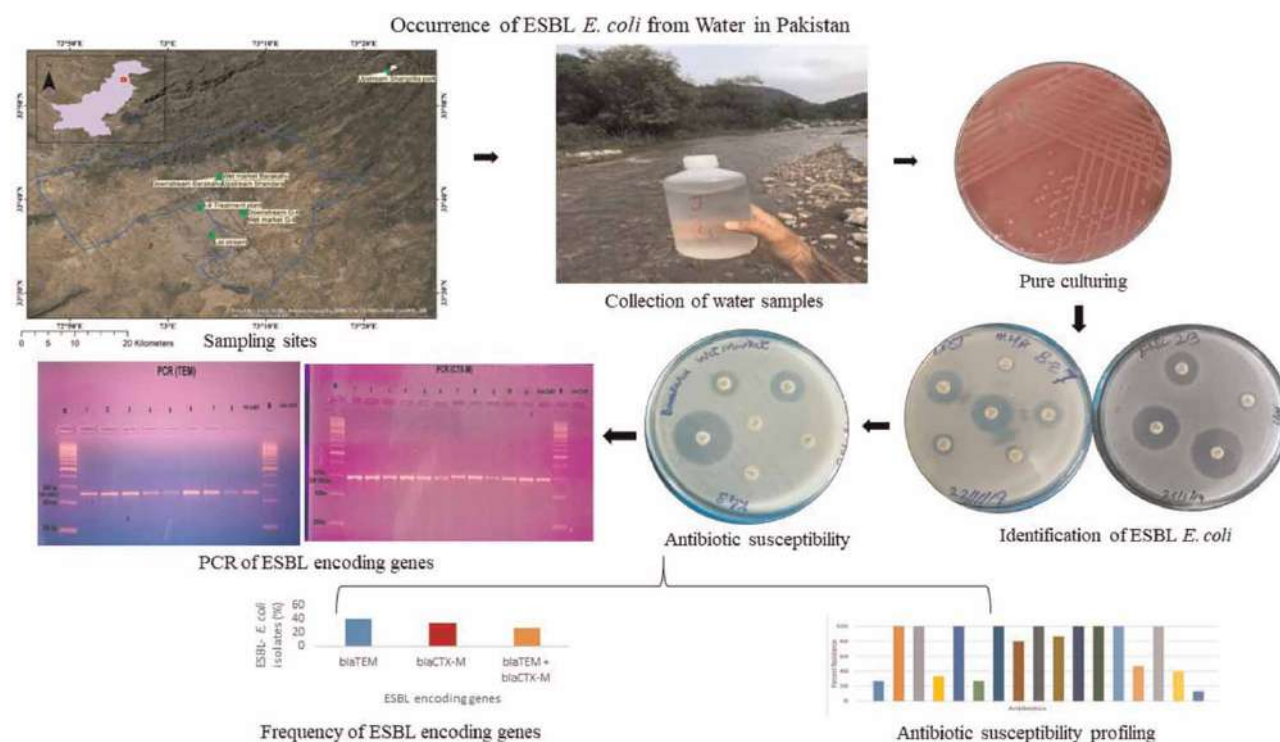
The goal of this study was to determine how surface and wastewater contribute to the contamination of the environment with an extended-spectrum β -lactamase-producing *Escherichia coli* (ESBL *E. coli*). Water samples ($n = 32$) were collected from eight different locations of Islamabad and processed for microbiological and molecular analyses of *E. coli* and ESBL *E. coli*. Antimicrobial susceptibility testing was carried out to determine the resistance pattern of the isolates. A total of 21 water samples were contaminated with *E. coli* and 15 isolates were identified as ESBL producers harboring *bla*_{TEM} (40%) and *bla*_{CTX-M} (33.33%) genes. Interestingly, all the ESBL *E. coli* isolates showed the least resistance against second-generation Cephalosporins compared to other generations. Moreover, the study showed that the aquatic environment is harboring multidrug-resistant *E. coli*; therefore, it may act as a source of transmission to humans. The recovery of ESBL *E. coli* isolates resistant to higher generation Cephalosporins, Monobactam, and Carbapenems from water samples indicated an alarming situation. Thus, there is an urgent need to treat water efficiently for microbial decontamination to minimize the transmission of antimicrobial-resistant (AMR) bacteria.

Key words: antibiotic resistance, ESBL *E. coli*, multidrug resistance, Pakistan, wastewater

HIGHLIGHTS

- This is the first study of its type from Pakistan which describes the occurrence of AMR bacteria in a water body starting from upstream to its subsequent downstreams.
- All (100%) of the ESBL *E. coli* isolates had multidrug resistance.
- The most commonly detected ESBL-encoding gene in ESBL *E. coli* isolates was *bla*_{TEM}.
- ESBL *E. coli* isolates resistant to last-resort antibiotics were recovered during the study.

GRAPHICAL ABSTRACT



INTRODUCTION

Antibiotic resistance especially against third-generation Cephalosporins and Carbapenems has become a threat for the global healthcare system. The most important resistance mechanism that undermines the efficacy of expanded spectrum Cephalosporins in the members of Enterobacteriaceae family is based on the production of plasmid-mediated enzymes, the extended-spectrum β -lactamases (ESBLs). These enzymes inactivate the said antibiotic compounds by disrupting their β -lactam rings (Zurfluh *et al.* 2013).

Escherichia coli (*E. coli*) is one of the most significant members of the family Enterobacteriaceae which have been affected with the emergence of ESBLs. Similarly, some strains of *E. coli* disseminating high levels of β -lactamases by the mutations of genes have been considered as an important group globally. *E. coli* is an outstanding indicator species to investigate the transmission of antimicrobial resistance through fecal contamination of water (Mohsin *et al.* 2017).

Antibiotic resistance has formerly been considered as a clinical problem, but recently natural ecosystems have been viewed as a crucial reservoir of antibiotic-resistant genes. The occurrence of antibiotic-resistant bacteria in aquatic environments is increasing consistently (Bouki *et al.* 2013). The water bodies, after receiving huge amounts of urban wastewater, hospital waste, and animal waste, represent the repository of diverse *E. coli*. These water bodies play a pivotal role in transferring and spreading of the resistance genes in public health significant pathogens and emerging unique resistance mechanism in them (Bajaj *et al.* 2015).

A recent study indicated *in vitro* transmission of ESBL-encoding multidrug-resistant (MDR) plasmids from ESBL *E. coli* recovered from water to susceptible *E. coli* making them antimicrobial-resistant (AMR) organisms (Gekenidis *et al.* 2020). Therefore, there is a possibility that ESBL *E. coli* isolates which are not resistant to antibiotics phenotypically but carry ESBL-encoding genes that may be involved in horizontal transmission of ESBL-encoding genes to other bacteria in the wastewater (Lenart-Boroń *et al.* 2020).

Only a few countrywide studies regarding the occurrence of ESBL *E. coli* in wastewater are reported; however, due to lack of any large-scale epidemiological study, the current understanding on environmental AMR reservoirs is very poor in Pakistan. Therefore, the current study was planned to (i) assess the occurrence of ESBL *E. coli* in wastewater of Islamabad Capital

Territory (ICT) of the country and (ii) to determine the percentage of most important β -lactamase (ESBL) genes, i.e. *bla*_{CTX-M}, *bla*_{SHV}, *bla*_{TEM} in the isolates obtained during the study and to estimate the current situation of antimicrobial resistance in *E. coli* isolates recovered during the study.

MATERIALS AND METHODS

Sample description

Water samples were collected from different locations of the Korang River and its tributaries (upstream Shangrilla Park, upstream Shahdara, downstream Baharakahu, and downstream G-6), Lai stream, wet markets, and wastewater treatment plant (WWTP) of the ICT. Sampling locations and their coordinates are shown in [Figure 1](#) and [Table 1](#).

Sampling was carried out fortnightly from May to October 2018. During this period, a total of 32 water samples were collected. Each sampling site was visited up to six times. The samples were taken in 1,000 ml sterilized glass bottles following the APHA, 2012 standard protocols. The samples were brought to the laboratory maintaining the cold chain and analyzed in the laboratory within 24 h.

Microbiological analysis for isolation of *E. coli*

Water samples were processed for microbiological analysis following APHA, 2012 guidelines. Briefly, samples were diluted in phosphate buffer saline (PBS) using a 10-fold serial dilution (up to 10^{-5}) and spread on tryptone bile X-glucuronide agar (TBX) and MacConkey agar supplemented with ceftriaxone (4 μ g/ml) and incubated at 37 °C for 24 h. Presumptive *E. coli* colonies were identified and confirmed using Gram's staining and biochemical tests including Indole, Methyl Red, Voges Proskauer, Citrate, and Triple sugar iron (TSI).

Identification of ESBL *E. coli*

Isolates identified as *E. coli* using Gram's staining and biochemical tests were analyzed for ESBL-producing characteristic using double-disk diffusion test (DDDT) and double-disk synergy test (DDST) ([Clinical and Laboratory Standards Institute \(CLSI\) 2013](#); [Garrec et al. 2013](#)).

Double-Disk Diffusion Test

Briefly, DDDT was performed by placing antibiotic disks of cefotaxime (with and without clavulanic acid) and ceftazidime (with and without clavulanic acid) on Muller Hinton agar (MHA) at a distance of 30 mm (center to center) from one another. The *E. coli* strain for which the zone of inhibition against the disks containing clavulanic acid was more in size (at least 5 mm) was considered an ESBL *E. coli* ([Clinical and Laboratory Standards Institute \(CLSI\) 2013](#)).

Double-Disk Synergy Test

Briefly, DDST was performed by placing ceftriaxone (30 μ g), ceftazidime (30 μ g), aztreonam (30 μ g), cefixime (30 μ g), and cefepime (30 μ g) disks on MHA at a distance of 30 mm center to center from amoxicillin-clavulanic acid (AMC 30) disk. The detection of ESBL strain was confirmed when any of the five antibiotic disks showed synergism toward the AMC disk, resulting in a typical 'keyhole-shaped' or 'ellipsis-shaped' zone ([Clinical and Laboratory Standards Institute \(CLSI\) 2013](#)).

Antibiotic resistance profile

Phenotypically confirmed ESBL *E. coli* isolates were tested for antibiotic susceptibility to 17 different antibiotics of veterinary and human clinical relevance using the disk diffusion method ([Kassim et al. 2016](#)). The antibiotics tested were amoxicillin, ampicillin, aztreonam, cephadrine, cefepime, cefixime, ceftazidime, ciprofloxacin, doxycycline, enrofloxacin, gentamicin, imipenem, lincomycin, neomycin, oxytetracycline, penicillin G, and streptomycin. Antimicrobial resistance against the tested disks was determined using CLSI breakpoints ([Clinical and Laboratory Standards Institute \(CLSI\) 2013](#)).

Molecular detection of ESBL-encoding genes

DNA was extracted from each of ESBL *E. coli* isolate using the boiling method ([Irshad et al. 2012](#)). Briefly, one ESBL *E. coli* colony was re-suspended in 200 μ l of autoclaved distilled water in a micro-centrifuge tube. This suspension was incubated at 95 °C in a heat block for 10 min and transferred to a freezer for 2 min for a clod shock. After centrifugation at 10,000 g for 5 min, the supernatant (containing DNA) was transferred to a fresh tube, stored at -20 °C, and was used as a template in PCR

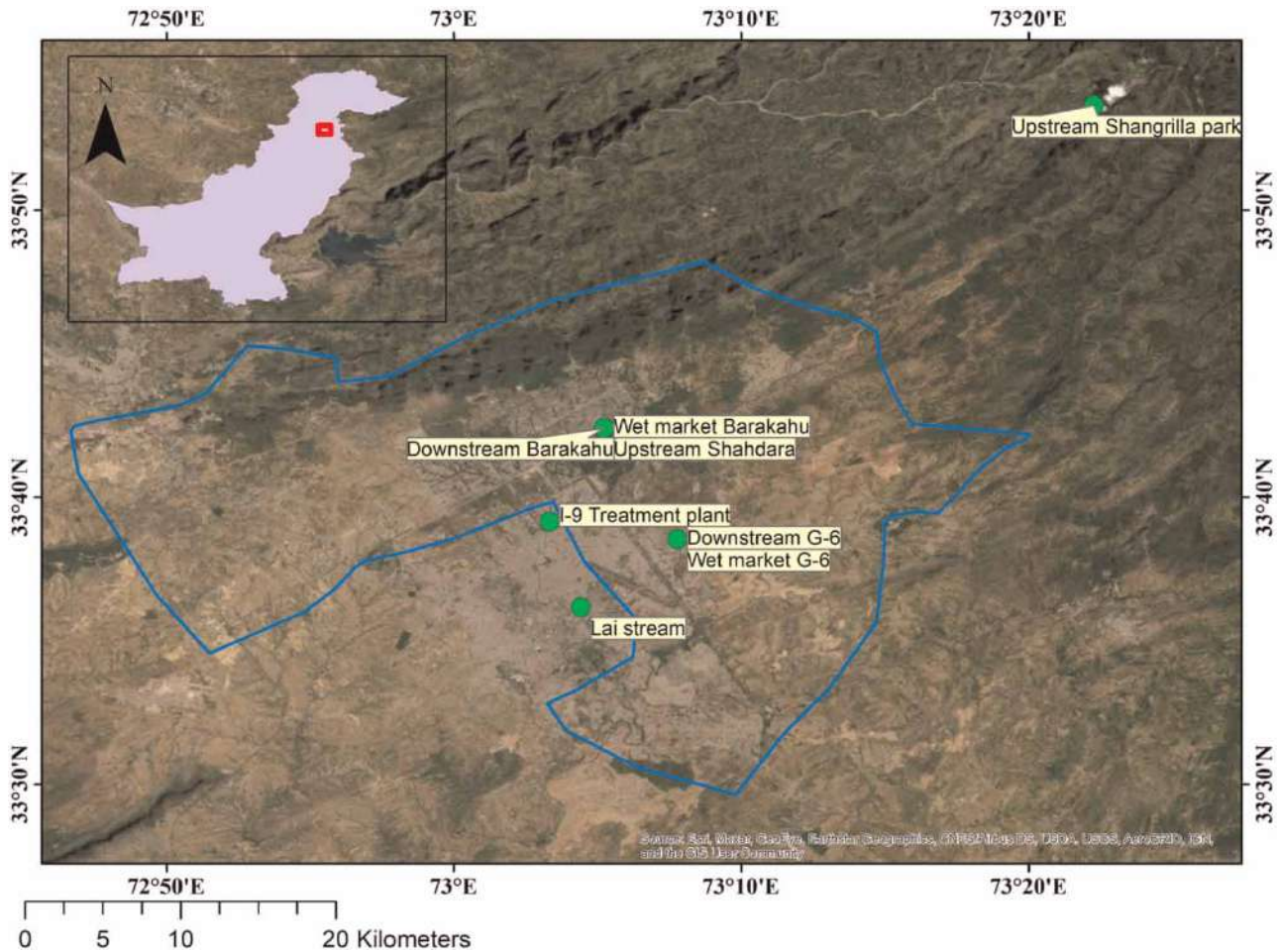


Figure 1 | Map showing water sampling locations in Islamabad Capital Territory.

reaction. The DNA was analyzed using PCR for the presence of *bla*_{TEM} (Gangoué-Piéboji *et al.* 2005), *bla*_{SHV} (Colom *et al.* 2003), and *bla*_{CTX-M} genes (Kaftandzieva *et al.* 2011). The PCR reaction mixture and thermal profile are given in Table 2.

RESULTS

Identification of *E. coli* isolates

In total 26 *E. coli* isolates were recovered from 32 water samples. The water samples from both upstreams (Shahdara and Shangrilla Park) were free from *E. coli* contamination. However, all of the water samples from wet markets and WWTP were heavily contaminated with *E. coli*. The details of *E. coli* contamination found in water samples from various sources are given in Table 3.

Occurrence of ESBL *E. coli* in wastewater

In total, 15 of 26 (57.69%) *E. coli* isolates recovered during the study period were found to be ESBL producers. Almost half of the *E. coli* isolates (13/23; 56.52%) recovered from downstream and wet markets were ESBL producers. This showed dissemination of AMR *E. coli* into environment. *E. coli* isolates recovered from WWTP and subsequent downstream (Lai stream) were also detected as ESBL producers (2/3; 66.67%). The details of isolation of ESBL *E. coli* are given in Table 3.

Antibiotic resistance patterns

The range of antimicrobial resistance (AMR) shown by ESBL *E. coli* isolates was 13.3–100%. All the ESBL *E. coli* isolates ($n = 15$) recovered during the study showed 100% resistance to streptomycin, neomycin (class Aminoglycosides), enrofloxacin (class Fluoroquinolones), lincomycin (class Lincosamides), ampicillin, penicillin G (class Penicillin), and oxytetracycline

Table 1 | GPS coordinates of locations from where the water samples were taken

Serial no.	Sample location	Coordinates
1	Upstream Shangrilla Park	33°53'38.7"N, 73°22'15.8"E
2	Downstream Baharakahu	33.7068°N, 73.0870°E
3	Wet market Baharakahu	33.7068°N, 73.0870°E
4	Upstream Shahdara	33.7068°N, 73.0870°E
5	Downstream G-6	33°38'32.6"N, 73°07'46.7"E
6	Wet market G-6	33°38'32.6"N, 73°07'46.7"E
7	I-9 treatment plant	33°39'09.7"N, 73°03'20.6"E
8	Lai stream	33°36'10.4"N, 73°04'25.1"E

Table 2 | PCR condition for detection of CTX, TEM, and SHV genes in ESBL *E. coli* isolates from water samples collected from eight different locations of Islamabad Capital Territory during May–October 2018

Gene	PCR conditions	PCR reaction volume	Amplicon size	Reference
CTX gene	1 cycle: 94 °C–5 min 35 cycles: 95 °C–30 s 54 °C–1 min 72 °C–1 min 1 cycle: 72 °C–8 min	2.5 µl of 10× PCR buffer 1.5 mM MgCl ₂ 0.2 mM of each dNTP (Thermoscientific, Waltham, USA) 1 µM of each primer (Macrogen, Seoul, South Korea) 1 unit of <i>Taq</i> DNA Polymerase (Thermoscientific, Waltham, USA) 5 µl of DNA Final volume of 25 µl with nuclease free water	588	Kaftandzieva <i>et al.</i> (2011)
TEM gene	1 cycle: 95 °C–5 min 35 cycles: 95 °C–30 s 56 °C–30 s 72 °C–1 min 1 cycle: 72 °C–10 min	2.5 µl of 10× PCR buffer 2 mM MgCl ₂ 0.2 mM of each dNTP (Thermoscientific, Waltham, USA) 2 µM of each primer (Macrogen, Seoul, South Korea) 1 unit of <i>Taq</i> DNA Polymerase (Thermoscientific, Waltham, USA) 5 µl of DNA Final volume of 25 µl with nuclease free water	465	Gangoué-Piéboji <i>et al.</i> (2005)
SHV gene	1 cycle: 94 °C–2 min 35 cycles: 94 °C–20 s 56 °C–60 s 72 °C–60 s 1 cycle: 72 °C–5 min	2.5 µl of 10× PCR buffer 1.5 mM MgCl ₂ 0.2 mM of each dNTP (Thermoscientific, Waltham, USA) 1 µM of each primer (Macrogen, Seoul, South Korea) 1 unit of <i>Taq</i> DNA Polymerase (Thermoscientific, Waltham, USA) 5 µl of DNA Final volume of 20 µl with nuclease-free water	392	Colom <i>et al.</i> (2003)

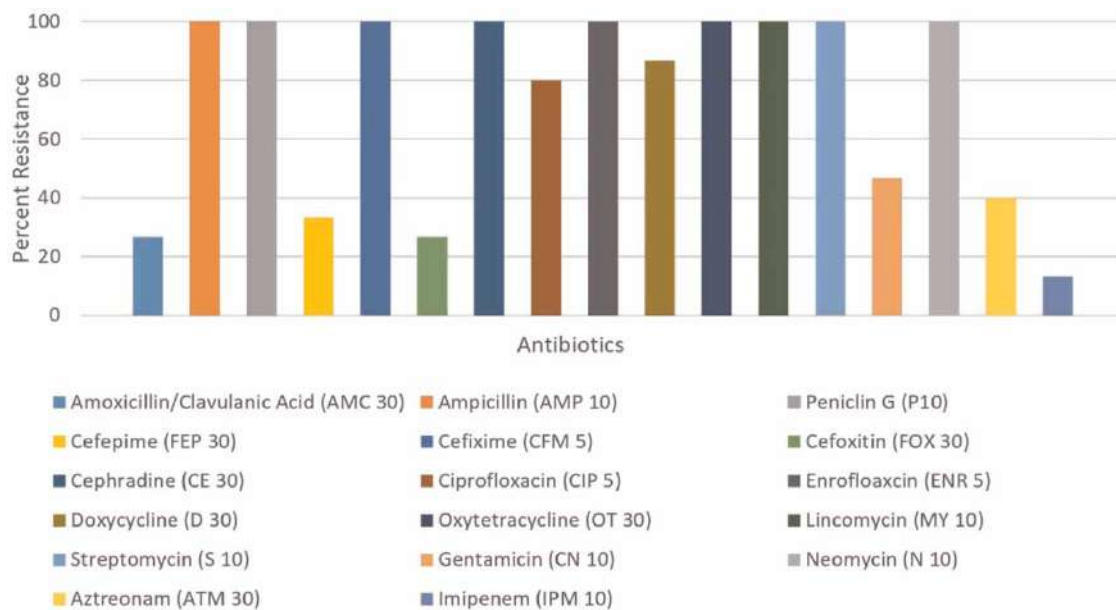
All the PCR reactions were carried out in PCR system Veriti (Applied Biosystems, Waltham, MA, USA).

(class Tetracycline). Moreover, the resistance of the isolates against doxycycline, ciprofloxacin, and gentamicin was 86.66, 80, and 46.66%, respectively. The results of Antimicrobial Susceptibility Testing (AST) against Cephalosporins were interesting as all isolates of ESBL *E. coli* showed least resistance against second-generation Cephalosporins (Cefoxitin; 26.66%), followed by fourth-generation Cephalosporins (Cefepime; 33.33%), first-generation Cephalosporins (Cephadrine; 100%), and third generation Cephalosporins (Cefixime; 100%), whereas they exhibited high susceptibility to Carbapenems (Imipenem; 86.67%) and amoxicillin with clavulanic acid (73.34%) (Figure 2).

Table 3 | Water samples ($n = 32$) were collected from eight different sites of Islamabad Capital Territory between May and October 2018

Serial no.	Sampling site	No. of samples collected	<i>E. coli</i> isolates recovered from samples	No. of <i>E. coli</i> isolates positive for ESBL
1	Upstream Shangrilla park	2	0/2 (0%)	0 (0%)
2	Downstream Baharakahu	6	5/6 (83.33%)	3/5 (60%)
3	Wet market Baharakahu	6	6/6 (100%)	3/6 (50%)
4	Upstream Shahdara	2	0/2 (0%)	0 (0%)
5	Downstream G-6	6	6/6 (100%)	3/6 (50%)
6	Wet market G-6	6	6/6 (100%)	4/6 (66.66%)
7	I-9 treatment plant	2	2/2 (100%)	1/2 (50%)
8	Lai stream	2	1/2 (50%)	1/1 (100%)

The samples were analyzed for the presence of *E. coli* and ESBL *E. coli* isolates.

**Figure 2** | Antibiotic susceptibility profile of ESBL *E. coli* isolates against 17 different antibiotics of veterinary and human clinical relevance using the disk diffusion method. The isolates were recovered from water samples collected from eight different locations of Islamabad Capital Territory during May–October 2018.

Frequency and diversity of ESBL-encoding genes

The most commonly detected ESBL-encoding gene in ESBL *E. coli* isolates was *bla*_{TEM} (06/15; 40%) followed by *bla*_{CTX-M} gene (05/15; 33.33%). The combination of *bla*_{TEM} and *bla*_{CTX-M} was observed in only 2 of 15 (13.33%) ESBL *E. coli* isolates. None of the ESBL *E. coli* isolates carried *bla*_{SHV} gene, whereas 4 of 15 (26.66%) ESBL *E. coli* isolates were negative for all three tested ESBL-encoding genes.

DISCUSSION

The water bodies have been considered as the key reservoir of antimicrobial-resistant bacteria. It has been demonstrated that fecal bacterial species, i.e. *E. coli*, *Salmonella*, and *Campylobacter* are being transmitted through water. This dissemination of the said fecal bacterial species with acquired AMR is a serious public health concern (Blaak *et al.* 2015). These AMR bacteria may be a source of transmission of resistance genes to other bacteria (Lenart-Boroń *et al.* 2020). In addition, these AMR bacteria may reach humans through direct contact with contaminated water or indirectly through surface water into food chain

and then to humans causing serious infections which are difficult to treat (Blaak *et al.* 2015). Moreover, *E. coli* species have been suggested for monitoring AMR in the environment (Mohsin *et al.* 2017). Therefore, keeping in mind the significance of this bacteria in the aquatic environment, the study was carried out to assess the burden of ESBL *E. coli* and its antibiotic resistance in surface and wastewater of ICT.

In total, 26 of 32 (81.25%) *E. coli* and 15 of 26 (57.69%) ESBL *E. coli* were recovered from water samples collected during this study. The water samples from upstream (Shangrilla Park and Shahdara) had no *E. coli* contamination. This may be due to limited or no anthropogenic activities in these areas. However, when the same water enters downstream of densely populated areas, i.e. Baharakahu and G-6, it was found to be contaminated with *E. coli* (91.66%) as these streams receive water from wet markets (poultry meat shops). It is a well-known fact that *E. coli* is a normal inhabitant of the gastrointestinal tract of poultry and other food animals (Seiffert *et al.* 2013). Therefore, excreta from these animals entering into the water of the streams may be the source of transmission of ESBL *E. coli* and other AMR bacteria into these streams. Furthermore, because ESBL *E. coli* are abundant in food animals, particularly broilers and calves, animal dung may also have a role in the transmission. Moreover, wild animal feces, such as those of birds, may introduce ESBL *E. coli* into surface water (Guenther *et al.* 2011).

All of the water samples from WWTP and its subsequent stream (Lai stream) were found polluted with AMR bacteria. Upon investigation from the WWTP management, it was revealed that wastewater is only treated for heavy metals and microbial decontamination procedure is not being carried out at present. Thus, there is an urgent need to treat water efficiently at treatment plants so that the transmission of AMR bacteria may be reduced.

High occurrence (57%) of ESBL *E. coli* from surface and wastewater was observed in this study which is consistent with previous studies of ESBL *E. coli* in India (64%) and Bangladesh (75%) (Haque *et al.* 2014; Johnson *et al.* 2020). This incidence of ESBL *E. coli* was in contrary to the incidence rates of ESBL *E. coli* from other sources in Pakistan, i.e. 38.80% from poultry farm environmental samples (Rahman *et al.* 2018) and 17.30% from migratory birds (Mohsin *et al.* 2017). However, the occurrence of ESBL *E. coli* in the current study was similar with that from humans (54%) in Pakistan (Hassan *et al.* 2011). Furthermore, we could not find any published data of ESBL *E. coli* from wastewater downstream and WWTP effluents in Pakistan to compare our results.

AST indicated that ESBL *E. coli* isolates were completely resistant to antibiotics which are considered as the first line of defense against infections. The AMR pattern of ESBL *E. coli* isolates observed against Cephalosporins during the study period was very interesting. As per general understanding, ESBL *E. coli* isolates should be least resistant to fourth-generation Cephalosporins followed by respective generations of the Cephalosporins. Nevertheless, the lowest resistance was seen against second-generation Cephalosporins (26.66%) followed by fourth-generation Cephalosporins (33.33%). The third- and first-generation Cephalosporins showed 100% resistance against the antibiotics tested in this study. One of the possible causes of this unusual resistance pattern could be the excessive and irrational usage of broad-spectrum and advanced generation Cephalosporins in the medical as well as in the veterinary/poultry sector.

Another finding of the study was the appearance of resistance in ESBL *E. coli* isolates against aztreonam (40%) and imipenem (13.33%). However, in a previous study carried out in Islamabad where *E. coli* isolates were recovered from meat samples obtained from retail meat showed high sensitivity (100%) against these antibiotics (Irshad *et al.* 2020). These antibiotics are costly and are deemed as the last choice antibiotics; therefore, they are hardly used in food animals (Irshad *et al.* 2020). The recovery of ESBL *E. coli* isolates resistant to aztreonam and imipenem from water samples in this study indicated that these streams of ICT may have received hospital waste effluents where these drugs are extensively used (Chantziaras *et al.* 2014).

The most prevalent gene found in 15 ESBL *E. coli* isolates from water samples during the current study was *bla*_{TEM} (40%, *n* = 6), followed by *bla*_{CTX-M} (33.33%, *n* = 5) which concurs previous report from Pakistan (Irfan *et al.* 2021). However, *bla*_{SHV} gene could not be detected in the tested ESBL *E. coli* isolates in this study. Similar occurrence of ESBL-encoding genes was reported in Germany except for *bla*_{SHV} gene (Savin *et al.* 2020). However, one study in contrary to the current study reported *bla*_{CTX-M} as the most prevalent gene in Pakistan (Abrar *et al.* 2018).

ESBL *E. coli* isolates negative for all the genes indicated that ESBL-encoding genes other than the three tested in the current study are also present in them. However, we tested for only three genes due to their widespread occurrence and clinical relevance in human population (Tissera & Lee 2013). These findings indicated that the ESBL *E. coli* isolates should be analyzed for other ESBL-encoding genes to understand the true distribution of ESBL-encoding genes in *E. coli* isolates.

The results of ESBL-encoding genes in the current study were quite interesting as the most prevalent gene was *bla*_{TEM} in the current study. Previously, *bla*_{CTX-M} is considered as the most predominant gene present in the region (Abrar *et al.* 2019). This

change in genomic epidemiology of ESBL *E. coli* suggested that the bacterial population in the environment may be facing selection pressure. The situation of AMR may become worse if it is true and this world can face a new pandemic in the form of AMR in near future.

The high occurrence of ESBL *E. coli* in water bodies of developing countries like Pakistan is a serious public health threat as antimicrobial usage is not being regulated in the medical as well as in the veterinary sector (Rahman *et al.* 2018). All (100%) of the ESBL *E. coli* isolates in the current study were resistant to multiple antimicrobial drugs simultaneously. This multiple drug resistance depicted the worrisome situation as the water bodies can be a source of transmission of these bacteria to humans. Moreover, a large diversity in the AMR pattern observed in this study is an indication of vulnerability and environmental pressure which bacterial species are facing in water streams (Bessa *et al.* 2014).

The study provided useful information about the occurrence of ESBL *E. coli* in water samples collected from different streams of ICT. Some of these isolates were resistant to more than two antibiotics indicating the presence of MDR bacteria in streams of ICT. However, ESBL *E. coli* isolates could not be recovered from the upstream water of Shangrilla and Shahdara where there is a limited anthropogenic activity. Although the study provided useful information about contamination of stream water of ICT with *E. coli* isolates but it was a small-scale study. Therefore, a large-scale study is required to completely understand the factors responsible for the occurrence of high level of *E. coli* in water of Korang River and its tributaries. The information would help in devising appropriate control strategies to reduce the prevalence of ESBL *E. coli* in environment and thus reaching humans.

CONCLUSION

This study showed that the aquatic environment is harboring MDR *E. coli*; and therefore may act as a source of transmission to humans. The recovery of ESBL *E. coli* isolates resistant to higher generation Cephalosporins, Monobactam, and Carbapenems from water samples indicated an alarming situation. Thus, there is an urgent need to treat water efficiently for microbial decontamination to minimize the transmission of AMR bacteria.

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CONFLICT OF INTEREST

There is no conflict of interest to declare.

AUTHORS' CONTRIBUTION

A.A. administrated the project, analyzed the samples and data, and wrote and edited the manuscript; T.A.U.R. collected the samples and data, and analyzed the samples; H.I. conceptualized, visualized, supervised, and analyzed the data, and wrote and edited the manuscript; M.A.S. analyzed the samples and data, and edited the manuscript; A.S. analyzed the samples and data; A.J. conceptualized and collected the samples; A.D. analyzed the samples.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

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